

Task 1: Install a Hypervisor

Steps:

1. Download VMware Workstation from the official VMware/Broadcom website.
2. Run the installer.
3. Follow the installation wizard.
4. Accept the license agreement.
5. Complete the installation.
6. Restart the computer if requested.
7. Open VMware Workstation.

Task 2: Create an Ubuntu Virtual Machine

Requirements

The VM should have at least:

- Guest OS: Ubuntu Linux
- RAM: 2 GB or more
- Storage: 20 GB or more
- CPU: 2 virtual processors recommended

Step 1: Download Ubuntu ISO

Download the Ubuntu Desktop ISO from the official Ubuntu website.

Official website:

<https://ubuntu.com/download/desktop>

Step 2: Create the VM

Open VMware Workstation.

Select:

- Create a New Virtual Machine

Choose:

- Typical (recommended)
- Select the downloaded Ubuntu ISO file.

Step 3: Configure the VM

Set the virtual machine name, for example:

- Ubuntu-Linux
- Set the storage location.

For disk size, select:

- 20 GB or more

For memory, configure:

- 2 GB minimum

If your computer has sufficient RAM, 4 GB is recommended for a smoother Ubuntu Desktop experience.

Step 4: Finish

Review the configuration and click:

- Finish

Task 3: Start Ubuntu VM

Start the virtual machine by clicking:

- Ubuntu will boot from the ISO.

Follow the Ubuntu installation wizard:

1. Select language.
2. Select keyboard layout.
3. Choose the installation type.
4. Create your Ubuntu username and password.
5. Complete the installation.
6. Restart the VM when requested.
7. Remove/detach the ISO if VMware asks you to do so.
8. Log in to Ubuntu.

Task 4: Type 1 vs Type 2 Hypervisors

Feature	Type 1 – Bare Metal	Type 2 – Hosted
Installation	Runs directly on physical	Runs on top of a host

	hardware	operating system
Performance	Generally provides better virtualization performance	Has additional overhead from the host OS
Common Usage	Servers and data centers	Personal computers, testing, and learning
Example	VMware ESXi	VMware Workstation

Three Differences

1. Architecture

- A Type 1 hypervisor runs directly on the physical hardware, while a Type 2 hypervisor runs as software inside an existing operating system.

2. Performance

- Type 1 hypervisors generally have lower virtualization overhead because they do not require a conventional host OS underneath them. Type 2 hypervisors have additional overhead because the host operating system sits between the hypervisor and physical hardware.

3. Usage

- Type 1 hypervisors are commonly used in enterprise servers and data centers, whereas Type 2 hypervisors are commonly used on desktop and laptop computers for testing, development, and education.

Real-World Examples

Type 1: VMware ESXi

Type 2: VMware Workstation